

**Mining Engineering department INDIAN INSTITUTE  
OF TECHNOLOGY KHARAGPUR**

**1. Name of the Academic Unit: Mining Engineering Department**

**2. Subject Name: Environmental Safety and Hazard Mitigation                      L-T-P:**  
**3-0-0                      Credits:3**

**3. Pre-requisites: None**

**4. Syllabus and reference books:**

**Environmental Safety and Hazard Mitigation**

Subsurface Environment: Atmospheric composition: Gaseous hazards from – seam gases, mine fires and explosions, diesel equipment, radioactive mines, underground blasting. Human respiratory safety and permissible norms for gases.

Spontaneous combustion: mechanism, susceptibility indices, factors affecting spontaneous combustion; detection and prevention of spontaneous heating. Fire hazards from other sources-causes and prevention; Fire control by direct and indirect methods, fire stoppings.

Subsurface explosions: Firedamp induced explosions-process, characteristics, risk estimation, flammability diagrams, and prevention. Coal dust induced explosions-process, characteristics, and prevention – stone-dust barriers, explosion proof stoppings.

Inundations: causes and prevention, precautions and techniques of approaching old workings; design of water dams, shaft dams, emergency bulk heads.

Rescue and Recovery practice in coal mines, reopening of sealed-off workings.

Mine illumination: physics of lighting, hazards due to low illumination, illumination norms, luminary sources, illumination design. Flame safety lamp.

Diesel particulate matter.

Airborne dust: characteristics, respiratory health issues, dust measurement, engineering duct control.

**Surface Environment:**

Air quality: Basics of atmospheric processes, influence of meteorological parameters on air quality, contaminant dispersion modeling, NAAQS

Surface mine dust: sources, measurement, control, health hazards

Water quality: basic ground water hydrology, water quality parameters, permissible norms for industrial effluents, acid mine drainage, heavy metal contamination, run-off control.

Land degradation: overburden dumps chemical and physical characteristics and plant growth potential, dump design for visual appeal, stability

Soil management: soil quality parameters, soil erosion in mines, soil remediation, reclamation and revegetation.

Noise: Fundamental of noise, noise sources in mines, permissible norms, propagation and attenuation, noise standards for mines, noise control.

Ground vibration: Transmission of blast induced vibrations through ground, structural safety, and estimation of PPV and dominant frequency, vibration standards.

#### **Reference Books:**

1. Ramlu, M.A.,1991, Mine Hazards and Mine Rescue, Oxford & IBH.
2. Vutukuri, V. S., & Lama, R. D.,1986, Environmental engineering in mines. Cambridge University Press.
3. Misra, G. B.,1986, Mine environment and ventilation. Oxford University Press.
4. Rao, M N & Rao, H V N, 1990, Air pollution, Publisher: New Delhi :Tata McGraw-Hill.
5. H S Pevy, D R Rowe and G Tchobanoglous,1985, Environmental Engineering McGraw-Hill Book, Singapore.
6. Mines Jr, R. O.,2014, Environmental engineering: principles and practice. John Wiley & Sons.
7. Gilbert M. Masters & Wendell P. Ela,2008, Introduction to Environmental Engineering and Science, 3rd edition. Prentice-Hall of India Private Limited, New Delhi.
8. Banerjee, S.C., 2000, Prevention and Combating Mine Fires, Oxford IBH
9. Vallero, D., 2014, Fundamentals of air pollution. Academic press

#### **5. Lecture-wise break-up:**

| <b>Sl. No.</b> | <b>Topic</b>                                                                                                                                                                                                                                                             | <b>No. of lectures</b> |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>1.</b>      | Atmospheric composition: Gaseous hazards from – seam gases, mine fires and explosions, diesel equipment, radioactive mines, underground blasting. Human respiratory safety and permissible norms for gases                                                               | <b>5</b>               |
| <b>2.</b>      | Spontaneous combustion: mechanism, susceptibility indices, factors affecting spontaneous combustion; detection and prevention of spontaneous heating. Fire hazards from other sources-causes and prevention; Fire control by direct and indirect methods, fire stoppings | <b>5</b>               |
| <b>3.</b>      | Subsurface explosions: Firedamp induced explosions-process, characteristics, risk estimation, flammability diagrams, and prevention. Coal dust induced explosions-process, characteristics, and prevention – stone-dust barriers, explosion proof stoppings              | <b>4</b>               |
| <b>4.</b>      | Inundations: causes and prevention, precautions and techniques of approaching old workings;                                                                                                                                                                              | <b>2</b>               |

|                              |                                                                                                                                                                                                                                                                  |           |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|                              | design of water dams, shaft dams, emergency bulk heads                                                                                                                                                                                                           |           |
| <b>5</b>                     | Mine illumination: physics of lighting, hazards due to low illumination, illumination norms, luminary sources, illumination design. Flame safety lamp.                                                                                                           | <b>2</b>  |
| <b>6</b>                     | Diesel particulate matter.<br>Airborne dust: characteristics, respiratory health issues, dust measurement, engineering duct control.                                                                                                                             | <b>3</b>  |
| <b>7</b>                     | Air quality: Basics of atmospheric processes, influence of meteorological parameters on air quality, contaminant dispersion modeling, NAAQS                                                                                                                      | <b>4</b>  |
| <b>8</b>                     | Water quality: basic ground water hydrology, water quality parameters, permissible norms for industrial effluents, acid mine drainage, heavy metal contamination, run-off control                                                                                | <b>4</b>  |
| <b>9</b>                     | Land degradation: overburden dumps chemical and physical characteristics and plant growth potential, dump design for visual appeal, stability<br>Soil management: soil quality parameters, soil erosion in mines, soil remediation, reclamation and revegetation | <b>4</b>  |
| <b>10</b>                    | Noise: Fundamental of noise, noise sources in mines, permissible norms, propagation and attenuation, noise standards for mines, noise control.                                                                                                                   | <b>3</b>  |
| <b>11</b>                    | Ground vibration: Transmission of blast induced vibrations through ground, structural safety, and estimation of PPV and dominant frequency, vibration standards.                                                                                                 | <b>3</b>  |
| <b>Total number of hours</b> |                                                                                                                                                                                                                                                                  | <b>39</b> |